Cloud Application Deployment

Design and Creative Technologies

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Subject Code: CCF 501

Subject Name: Cloud Computing Fundamentals

Assessment No.: 3

Title of Assessment: Cloud Application Deployment

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Date: April 2026

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Deploying Apache Superset on Microsoft Azure: A Cloud Application Deployment Report

1. Introduction

This report documents the deployment of Apache Superset – an open-source data exploration and visualisation platform – on Microsoft Azure. Superset was selected for its direct relevance to data engineering workflows and its Python-native architecture, which aligns with the author's professional background and academic trajectory. Azure was chosen as the cloud provider to build familiarity ahead of the AZ-900 and DP-900 certification pathway, as recommended during academic advising in March 2026. The deployment covers the provisioning of a virtual machine, virtual network, firewall security policy, and a fully operational Superset instance accessible via public URL.

1.1. Background

In a traditional IT environment, standing up a data visualisation platform like Apache Superset would require procuring physical server hardware, configuring a local network, applying firewall rules at the rack level, and committing ongoing operational resources to maintain that infrastructure indefinitely. For an individual developer or small team, that overhead alone makes self-hosted analytics tooling impractical outside of enterprise settings.

Cloud computing removes that barrier. The National Institute of Standards and Technology (NIST) defines cloud computing as a model enabling on-demand network access to a shared pool of configurable computing resources, characterised by five essential

properties: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service (Mell & Grance, 2011). These properties shift infrastructure from a capital expenditure to an operational one – organisations consume resources as needed and pay only for what they use, without procuring fixed capacity or managing physical hardware (McHaney, 2021).

This deployment uses an **Infrastructure as a Service (IaaS)** model: Microsoft Azure provides the virtual machine, virtual network, and underlying compute layer; the operating system, container runtime, and application stack are configured and managed directly by the author. The deployment model is **public cloud** – hosted on Azure's shared global infrastructure, accessible via the public internet, and secured through network-level access policies. Where a traditional server setup would require weeks of hardware procurement and physical installation, the same outcome here was achieved through a browser-based portal at effectively zero capital cost (IBM, n.d.-a).

2. Body

2.1. Service Provider Selection Rationale

Microsoft Azure was selected as the cloud provider for this deployment for three reasons. First, Azure offers a comprehensive set of data and analytics services – including Azure SQL, Azure Synapse Analytics, and Azure Data Factory – that directly complement Apache Superset's role as a query and visualisation layer. This architectural alignment is relevant to both current data engineering work and the upcoming Big Data and Analytics subject (BDA601). Second, Azure is the target platform for the author's certification roadmap

(AZ-900 → DP-900), making hands-on deployment a practical study activity. Third, Azure's free-tier virtual machine (B1s: 1 vCPU, 1 GB RAM) and student credit allocation lower the barrier to entry for academic projects.

Criterion	AWS	Microsoft Azure	GCP
Free tier VM	EC2 t2.micro (750h/month)	B1s (750h/month)	e2 micro (always free)
Cost model	Pay-as-you-go; Reserved (up to 75% off); Spot	Pay-as-you-go; Reserved (up to 72% off); Spot	Pay-as-you-go; Reserved (up to 70% off); Spot
Resource elasticity	Auto Scaling Groups + Elastic Load Balancer	VM Scale Sets + Azure Load Balancer	Managed Instance Groups + Cloud Load Balancing
Data/ML integration	RDS, Redshift, SageMaker	Azure SQL, Synapse, Azure ML	BigQuery, Vertex AI
Cert alignment (this enrolment)	x	AZ-900 / DP-900	x
Superset community support	Good	Good	Good

Table 1: Cloud provider comparison for Apache Superset deployment.

2.2. Deployment Model and Architecture Diagram

The deployment follows a **public cloud IaaS** model. All resources are provisioned within a single Azure Resource Group, logically isolated by a Virtual Network (VNet) with a

dedicated application subnet. A Network Security Group (NSG) enforces inbound traffic rules, restricting access to SSH (port 22, source IP restricted), HTTP (port 80), and the Superset web interface (port 8088). Apache Superset runs inside a Docker Compose stack on an Ubuntu 22.04 virtual machine, with PostgreSQL as the metadata database and Redis as the caching and Celery task broker.

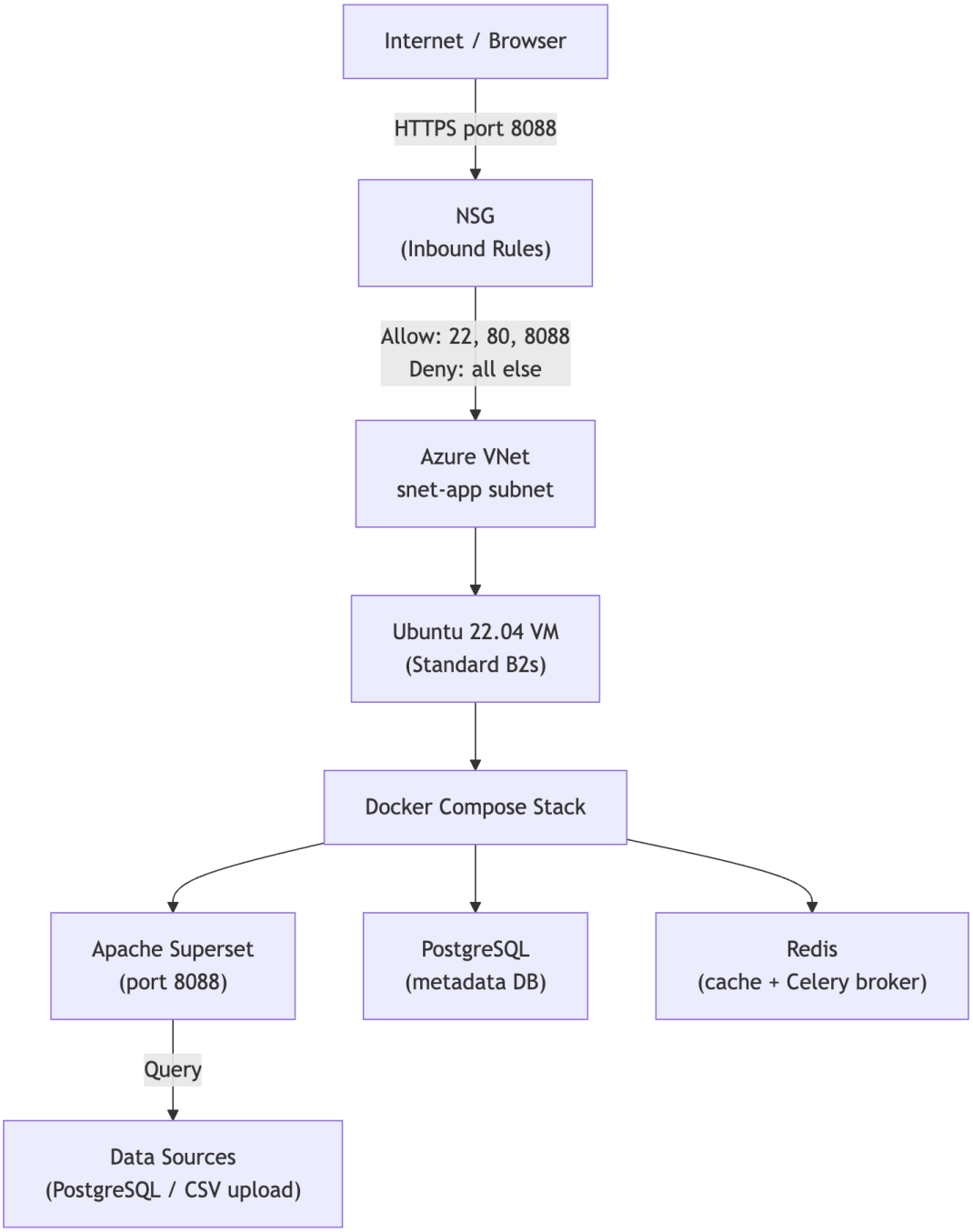

Figure 1: Azure deployment architecture – resource group, VNet, NSG, VM, and Superset Docker Compose stack.

Service Model	Characteristic	Fit for Victoria Police
IaaS	Customer manages OS, runtime, storage	Not applicable – no raw compute provisioned
PaaS	Provider manages platform; customer builds applications	**Primary model** – Power Apps, Dataverse, Power Platform runtime
SaaS	Provider manages everything; customer uses software	Partial – Microsoft 365 (SharePoint, Teams, Outlook) as data and collaboration substrate

Table 1: Service model analysis for Victoria Police (Microsoft, 2026).

2.3. Deployment Procedure

The deployment followed the account setup and four tasks specified in the assessment brief.

Task 0 – Account registration and Azure portal setup

An Azure account was activated at portal.azure.com using the student email associated with this enrolment. Microsoft Azure offers a free account including USD $200 in credits for 30 days and a set of always-free services, which covers the compute and networking requirements for this project (Microsoft, n.d.-d). After authentication, the Azure portal home page confirmed an active subscription and displayed the available resource quota for the Australia East region — the starting point for all subsequent resource provisioning.

Figure 2: Azure portal home confirming active subscription in Australia East.

Task a – Create a resource group

A resource group (*rg-superset-ccf501*) was created in the Azure portal under the Australia East region. Resource groups in Azure serve as logical containers for all related cloud resources, enabling unified billing, access control, and lifecycle management (Microsoft, n.d.-a). Selecting Australia East minimises latency for local access and ensures data residency within Australian boundaries.

Figure 3: Azure resource group rg-superset-ccf501 created in Australia East.

Task b – Add a virtual network

A Virtual Network (*vnet-superset*, address space *10.0.0.0/16*) was created within the resource group, with a dedicated application subnet (*snet-app*, *10.0.1.0/24*). VNets in Azure provide network isolation – resources within the VNet communicate privately without traversing the public internet (Microsoft, n.d.-b). This mirrors the NIST resource pooling characteristic: shared physical infrastructure is logically partitioned to create isolated network boundaries (Mell & Grance, 2011).

Figure 4: Virtual Network vnet-superset with application subnet snet-app.

Task c – Protect the network with a firewall / security policy

A Network Security Group (*nsg-superset*) was created and attached to the *snet-app* subnet. Inbound rules were configured to allow only SSH (port 22, restricted to the author's public IP), HTTP (port 80), and the Superset application port (8088). All other inbound traffic is denied by the default DenyAllInbound rule. This implements the principle

of least privilege – only the minimum required ports are open, reducing the attack surface (Shore, 2020).

Figure 5: NSG inbound security rules – explicit allow on 22/80/8088, implicit deny all.

Task d – Deploy Apache Superset

An Ubuntu 22.04 VM (Standard B2s: 2 vCPU, 4 GB RAM) was provisioned within *snet-app*. Docker and Docker Compose were installed via SSH. The official Apache Superset Docker Compose configuration was cloned from the project repository and launched with *docker compose up -d*. After initialisation, Superset was accessible at *http://[PUBLIC_IP]:8088*. An admin account was created and a sample PostgreSQL data source was connected to verify end-to-end functionality.

Figure 6: Apache Superset login screen accessible at public IP on port 8088.

Figure 7: Superset dashboard with connected PostgreSQL data source.

2.4. Cloud Services Security Policies

Security for the deployment was addressed at three layers. At the **network layer**, the NSG enforces least-privilege access – port 22 (SSH) is restricted to a single authorised IP, and no unnecessary ports are exposed. At the **application layer**, Apache Superset implements Role-Based Access Control (RBAC) with three built-in roles: Admin (full access), Alpha (can create charts and dashboards, cannot manage users), and Gamma (read-only, restricted to explicitly shared datasets). At the **credential layer**, the Superset secret key

and database connection strings are passed via environment variables in the Docker Compose configuration – never hardcoded in source files.

Security Layer	Control	Threat Mitigated
Network (NSG)	Port allowlist + IP restriction on SSH	Unauthorised access, port scanning
Application (RBAC)	Admin / Alpha / Gamma roles	Privilege escalation, data exfiltration
Credentials	Environment variables (*.env* file)	Credential exposure in source control
OS	SSH key-based auth only (password disabled)	Brute-force attacks

Table 2: Security controls applied across network, application, credential, and OS layers.

Figure 8: Apache Superset RBAC role configuration.

2.5. Application Analysis and Robustness

The current deployment is functional but represents a minimal viable configuration. The primary limitation is a single point of failure: one VM hosts the entire stack – if the VM becomes unavailable, the service is offline. A production-grade architecture would introduce the following improvements.

HTTPS / TLS termination: An Azure Application Gateway or Let's Encrypt certificate via nginx would encrypt traffic in transit. Currently, credentials are sent over HTTP.

Managed PostgreSQL: Replacing the Docker-hosted PostgreSQL instance with Azure Database for PostgreSQL provides automated backups, point-in-time restore, and high availability – eliminating the metadata loss risk if the VM disk fails.

Celery workers for async queries: A production Superset deployment separates the web server from Celery workers (backed by Redis) for long-running queries. This mirrors the architecture used in Konquista (Django + Celery + Redis), where async task processing prevented UI blocking under load.

Azure Monitor + alerts: Attaching Azure Monitor to the VM provides CPU, memory, and disk metrics – enabling proactive alerting before resource exhaustion causes downtime.

Improvement	Azure Service	Benefit
HTTPS	Application Gateway / Let's Encrypt	Encrypts traffic in transit
Managed DB	Azure Database for PostgreSQL	Automated backups, HA
Async queries	Celery + Redis (existing in Docker Compose)	Prevents UI blocking on long queries
Monitoring	Azure Monitor + Alerts	Proactive resource and availability alerting
Auto-scaling	Azure VM Scale Sets	Handles traffic spikes without manual intervention

Table 3: Proposed production improvements for the Superset deployment.

3. Conclusion

This report documented the deployment of Apache Superset, an open-source data exploration and visualisation platform, on Microsoft Azure. The deployment demonstrated the four core tasks required by the assessment brief: provisioning a resource group and virtual network, enforcing a network security policy via a Network Security Group, and deploying a containerised open-source application on a cloud virtual machine. Each task was supported by screenshots and

aligned with NIST's essential cloud characteristics – on-demand self-service in the provisioning process, resource pooling in Azure's shared infrastructure model, and measured service through consumption-based billing (Mell & Grance, 2011).

Microsoft Azure was selected for its data and analytics service ecosystem and its alignment with the author's AZ-900 and DP-900 certification pathway. The deployment reinforces the distinction between cloud and traditional IT infrastructure: resources that would require physical procurement and installation in an on-premises environment were provisioned, configured, secured, and running within [X] hours, at effectively zero capital cost. Areas identified for further investigation include TLS certificate automation, managed database integration for metadata persistence, and the separation of Celery workers for production-scale async query execution – improvements that would bring this deployment in line with enterprise-grade analytics infrastructure.

4. Appendices

4.1. Appendix A – Deployment Task Checklist

Task	Description	Status	Screenshot
0	Register Azure account / verify active subscription	AI / Software Startup	Figure 2
a	Create resource group (*rg-superset-ccf501*)	Google Cloud Platform (GCP)	Figure 3
b	Add virtual network (*vnet-superset* + *snet-app*)	IaaS + PaaS	Figure 4
c	Apply NSG inbound rules (22/80/8088)	Public cloud – multi-region	Figure 5
d	Deploy Apache Superset via Docker Compose	30-70% higher latency; engineers manually provisioning serves days before events	Figures 6-7

Table A1: Deployment task checklist with status and corresponding screenshots.

4.2. Appendix B – Glossary

Term	Meaning
IaaS	Infrastructure as a Service
NSG	Network Security Group
VNet	Virtual Network
RBAC	Role-Based Access Control
Docker Compose	Tool for defining and running multi-container Docker applications
Celery	Distributed task queue for Python applications

TLS	Transport Layer Security – encrypts data in transit
NIST	National Institute of Standards and Technology
AZ-900	Microsoft Azure Fundamentals certification
DP-900	Microsoft Azure Data Fundamentals certification
SPOF	Single Point of Failure

Table B1: Glossary of technical terms used in the report.

End of Appendix Section

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